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Methods and Algorithms for Molecular Docking– Based Drug Design and Discovery

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Chapter 9

Scoring Functions of Protein–Ligand Interactions

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ABSTRACT

Scoring function of protein-ligand interactions is used to recognize the “native” binding pose of a ligand on the protein and to predict the binding affinity, so that the active small molecules can be discriminated from the non-active ones. Scoring function is widely used in computationally molecular docking and structure-based drug discovery. The development and improvement of scoring functions have broad implications in pharmaceutical industry and academic research. During the past three decades, much progress have been made in methodology and accuracy for scoring functions, and many successful cases have be witnessed in virtual database screening. In this chapter, the authors introduced the basic types of scoring functions and their derivations, the commonly-used evaluation methods and benchmarks, as well as the underlying challenges and current solutions. Finally, the authors discussed the promising directions to improve and develop scoring functions for future molecular docking-based drug discovery.

INTRODUCTION

The experiment-based discovery and development of novel drugs are highly complex, time-consuming and expensive (Dickson, 2004). As the number of determined three-dimensional protein structures grows and the power of high-performance computing improves, structure-based drug design and discovery is gaining prominence to identify lead compounds. This field has experienced a rapid growth in academia and pharmaceutical industry. Molecular docking is the most popular computational method employed in structure-based drug design and discovery (Kitchen, 2004; Meng, 2011; Bello, 2013).

Molecular docking mimics the recognition process in which a small molecule (ligand) translates, rotates, and twists thoroughly in the active site of a macromolecule (normally protein), with the goal of finding the most favorable binding mode measured by a scoring function of the binding interactions.

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Thus, the protein-ligand docking method can be described as a combination of a search algorithm and a scoring function. The search algorithm generates a number of possible poses that fit the ligand into the binding pocket of the protein. The scoring function scores each pose and then ranks the different poses that are generated by the search algorithm.

In the past three decades, multiple search algorithms were employed in the molecular docking and different types of scoring functions were developed (Dias, 2008; Guo, 2014; Liu, 2015). Various docking programs based on different search algorithms and scoring functions were developed to perform molecular docking studies and applications in structure-based drug design and discovery (Plewczynski, 2011; Sousa, 2013). There have been a mounting number of successful stories with docking-based virtual screening (Tuccinardi, 2009; Villoutreix, 2009; Danishuddin, 2015). Those successes are the outcome of the concerted applications from sampling algorithms, scoring functions as well as filtering criteria. In this chapter, the authors would like to focus on the introduction of scoring functions.

BACKGROUND

Experimentally, the binding affinity (free energy) ($\Delta G_{\text{Bind}} = G_{\text{Complex}} - G_{\text{Protein}} - G_{\text{Ligand}}$) of a protein-ligand complex can be given by the determined dissociation constant (K_d) with the equation $\Delta G_{\text{Bind}} = RT \ln K_d$, where R is the ideal gas constant, and T is the temperature. Computationally, there are a wide variety of different methods capable of predicting the binding affinity of protein-ligand complex. Relatively accurate binding methods, such as free energy perturbation (FEP), thermodynamic integration (TI), linear interaction energy (LIE), Molecular mechanics Poisson-Boltzmann (or Generalized Born) surface area (MM-PB(GB)SA) continuum solvation models (Kollman, 1993; Aqvist, 1994; Gutiérrez-de, 2012; Kollman, 2000), require large computational cost, especially in the computational virtual screening of small molecule databases. Instead, the scoring functions adopt various assumptions and simplifications with approximate mathematical methods to estimate the binding free energy between the protein and the ligand; they are fast in calculating the protein-ligand interactions. Thus, the accuracy is the main task of developers in improving and developing scoring functions.

The accuracy of a scoring function can be evaluated in accordance with its applications in the structure-based drug design and discovery (Huang, 2010; Li, 2014). The first one refers to the ability to rank the computationally generated docking poses, and identify the top ranked one as the “native” binding pose which should be close to experimentally determined structure. The second one refers to the ability to estimate the binding affinities which should be proportional to the experimentally determined binding affinities. The third one refers to the ability to discriminate the true binders against inactive ones in the virtual database screening according to computational binding scores. With the appropriate evaluation methods, these three kinds of abilities can be validated by testing on the protein-ligand benchmarks or with blind assessment exercises.

Scoring functions are either implemented in docking programs or stand-alone in scoring and ranking the generated poses or ligands. So far, the scoring functions have grown into a large diverse family with more than one hundred, whereas, many scoring functions employ common methodologies. Therefore, the scoring functions can be roughly grouped into four main categories: force-field-based, empirical, knowledge-based, and machine-learning-based scoring functions (Liu, 2015). In addition, to take the advantages and balance the deficiencies of different scoring functions, the consensus scoring technique has been introduced to improve the probability of finding correct solutions by combining the scores from multiple scoring functions.

Though the molecular docking is playing an increasing role in structure-based drug design and discovery, important challenges are still existed, such as water molecules in binding site, solvation effect, protein flexibility, treatment of entropy, covalent docking, metal ions, and binding specificity (Yan, 2013; Yuriev, 2013; Brás, 2014). Some challenges lie in the side of scoring functions and common to all types of scoring functions. By addressing these obstacles, many attempts have been made to the scoring functions to increase the robustness of reliability and accuracy in their applications, which have paved the way for future directions of improving and developing new scoring functions.

SCORING FUNCTIONS

Classes of Scoring Functions

1. Force-Field-Based Scoring Function

Since the pioneer works by Martin Karplus and coworkers employed molecular mechanics force fields to model biological macromolecules (Karplus, 1976; Brooks, 1983), force-field-based scoring functions have been used for more than two decades to compute the binding affinity between the protein and the ligand (Meng, 1992; Kollman, 1993; Goodsell, 1996; Makino, 1997; Jones, 1997; Kramer, 1999; Yin, 2008). Force-field-based scoring functions generally quantify the sum of two energies, the protein-ligand interaction energy and the internal energy of the ligand. The functional form of the internal ligand energy is very similar to the protein-ligand interaction energy. The energies are normally composed of a van der Waals and an electrostatic energy terms. The general functional form is like

$$S = E_{PL} + E_L \quad (1)$$

where E_{PL} represents the energy between the protein and ligand and E_L represents the internal energy of the ligand. Different force-field-based scoring functions have similar energy forms with a few different extension energy terms as shown in Table 1. The van der Waals energy term can be given by the Lennard-Jones potential function, the parameters of the Lennard-Jones potential vary by using 12-6 or 8-4 Lennard-Jones potential to adjust the “hardness” of the potential. The electrostatic term is accounted for by a Coulombic formulation with a distance-dependent dielectric function. The effect of solvent is implicitly considered by introducing a simple distance dependent dielectric constant in the Coulombic formulation, which lessens the contribution from charge-charge interactions by mimicking the screening effect. The extensions of general force-field-based scoring functions include a torsional entropy term for ligands and the inclusion of explicit protein-ligand hydrogen-bonding terms. The torsional entropy term is to account for the entropy associated with the rotatable bonds of the ligands. For example, immobilization of a rotatable bond on binding leads to loss of its torsional (or rotational) entropy. The hydrogen-bonding term is modeled to increase the potential of specific ligand binding with noticeably different ways. For example, G-Score (Kramer, 1999) includes different hydrogen-bonding terms depending on the nature and geometry of the interaction while AutoDock represents all of the hydrogen bonds by a 12-10 Lennard-Jones potential with an angular weighting factor.

Scoring Functions of Protein-Ligand Interactions

Table 1. Examples of force-field-based scoring functions

Name	Functional Form
AutoDock4.2 (Morris, 2009)	$S = W_{vdw} \sum_{i,j} \left(\frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^6} \right) + W_{hb} \sum_{i,j} E(t) \left(\frac{C_{ij}}{r_{ij}^{12}} - \frac{D_{ij}}{r_{ij}^{10}} \right) + W_{ele} \sum_{i,j} \frac{q_i q_j}{\epsilon(r_{ij}) r_{ij}}$ $+ W_{sol} \sum_{i,j} (S_i V_j + S_j V_i) e^{(-r_{ij}^2 / 2\sigma^2)}$
Dock (Makino, 1997)	$S = \sum_{i,j} \left[\left(\frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^6} \right) + 332.0 \frac{q_i q_j}{\epsilon(r_{ij}) r_{ij}} \right]$
GoldScore (Jones, 1997)	$S = \sum_{i,j} \left[\left(\frac{A_{ij}}{r_{ij}^a} - \frac{B_{ij}}{r_{ij}^b} \right) + 332.0 \frac{q_i q_j}{\epsilon(r_{ij}) r_{ij}} \right] + (E_{hb})_{optional}$
D-Score (Meng, 1992)	$S = \sum_{i,j} \left[\left(\frac{A_{ij}}{r_{ij}^{12}} - \frac{B_{ij}}{r_{ij}^6} \right) + 332.0 \frac{q_i q_j}{\epsilon(r_{ij}) r_{ij}} \right]$
G-Score (Jones, 1997)	$S = \sum_{i,j} \left[\left(\frac{A_{ij}}{r_{ij}^a} - \frac{B_{ij}}{r_{ij}^b} \right) + 332.0 \frac{q_i q_j}{\epsilon(r_{ij}) r_{ij}} \right] + \sum_{lig} \frac{1}{2} V \left[1 + \frac{n}{ n } \cos(n \omega) \right] + E_{hb}$
Atoms i and j are either from the protein and ligand respectively or both from the ligand; A, B, C and D are van der Waals parameters for given atoms, C and D are specialized for hydrogen bonds with oxygen and nitrogen; Deferent W are the weighting constants for the energy terms; E(t) is the angle weighting factor at angel t from ideal hydrogen bonding geometry; ϵ is the dielectric constant; E_{hb} is the hydrogen bonding term; a and b means that the protein-ligand interaction energy and the internal energy would take different Lenard-Jones parameters; n, w and V are the parameters for the torsional energy in the ligand. For more details, readers are suggested to look into the related citations.	

Force-field-based scoring functions were primarily formulated to model enthalpy gas-phase contributions to the energetics. Thus, the major limitations of force-field-based scoring functions are the lack of solvation and entropic terms which are important contributors to the binding free energy between the protein and the ligand. This is in part corrected through the inclusion of additional terms. For example, G-Score includes a torsional entropy term to estimate the conformational entropy loss of ligand upon binding. A straightforward way to account for the solvent effect is to treat water molecules explicitly such as free energy perturbation (FEP) and thermodynamic integration (TI). Alternatively, implicit solvent models include Poisson-Boltzmann (PB) or Generalized Born (GB) continuum solvation models (Kollman, 2000), were developed for scoring use in molecular docking. These methods compute energies as ensemble averages in an attempt to incorporate more physics at the expense of significantly increased computational cost. In practice, these methods are often used in post-scoring of docking programs.

2. Empirical Scoring Function

The empirical scoring function was first proposed by Böhm in 1992 (Böhm, 1992). An empirical scoring function computes the fitness of protein-ligand binding by summing up the contributions of a number of individual terms. The functional form of empirical scoring functions is often simpler than force-field-based scoring functions, although many of the individual contributing terms have counterparts in the force-field molecular mechanics terms. Generally, the empirical scoring functions decompose the overall binding free energy into several energetic terms as

$$S = \sum_i W_i E_i \quad (2)$$

The coefficients W_i for various terms E_i are obtained from regression analysis using experimentally determined binding free energies and sometimes structural information. Representative examples of empirical scoring functions are given in Table 2. Though the basic expressions are similar between different empirical scoring functions, the energetic terms have rather different implementations. For

Table 2. Examples of empirical scoring functions

Name	Functional Form
LUDI (Böhm, 1992)	$S = \Delta G_{hb} \sum_{hb} f(\Delta R, \Delta \alpha) + \Delta G_{ion} \sum_{ion} f(\Delta R, \Delta \alpha) + \Delta G_{hp} \sum_{hp} A_{hp} + \Delta G_{rot} N_{rot} + \Delta G_0$
ChemScore (Eldridge, 1997)	$S = \Delta G_{hb} \sum_{hb} f(\Delta R, \Delta \alpha) + \Delta G_{met} \sum_{met} f(\Delta R, \Delta \alpha) + \Delta G_{lipo} \sum_{lipo} f(\Delta R) + \Delta G_{rot} \sum_{rot} f(p_{nl}, p'_{nl}) + \Delta G_0$
F-Score (Rarey, 1996)	$S = \Delta G_{hb} \sum_{hb} f(\Delta R, \Delta \alpha) + \Delta G_{ion} \sum_{ion} f(\Delta R, \Delta \alpha) + \Delta G_{arom} \sum_{arom} f(\Delta R) + \Delta G_{con} \sum_{con} f(\Delta R, \Delta \alpha) + \Delta G_{rot} N_{rot} + \Delta G_0$
GlideScore (Wang, 2002)	$S = C_{lipo-lipo} \sum f(r_{lr}) + C_{hb-neut-neut} \sum g(\Delta r) h(\Delta \alpha) + C_{hb-neut-ch} \sum g(\Delta r) h(\Delta \alpha) + C_{hb-ch-ch} \sum g(\Delta r) h(\Delta \alpha) + C_{max-met-ion} \sum f(r_{lr}) + C_{rot} H_{rot} + C_{pol-hp} V_{pol-hp} + C_{coul} E_{coul} + C_{vdw} E_{vdw} + E_{sol}$
The scoring above uses energy terms including hydrogen bonding (hb), ionic (ion), hydrophobic (hp), ligand rotational entropy (rotor), contact (con), lipophilic (lipo), metal (met) Coulomb (coul), van der Waals (vdw) and solvation components. The scoring functions differ in energy terms and corresponding regression coefficients. Different ΔG are obtained by multiple linear regressions, ΔG_0 is a regression constant, and the energy term is calculated with a function f or g or h which depends on the angular and/or a distance. GlideScore modifies and extends the ChemScore by classifying hydrogen bonding term into several weighted components depending on whether the donor and the acceptor are neutral (neut) or charged (ch), and incorporating the Coulomb (coul) and van der Waals (vdw) as well as solvation (sol) component. For more details, the readers are suggested to look into the related citations.	

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instance, LUDI function computes hydrophobic contributions based on the molecular surface area; Glide XP and Fresno (Rognan, 1999) functions include the solvation term to account for the desolvation effect; F-Score function takes into account the interaction between aromatic groups.

Both force-field-based and empirical scoring functions decompose protein-ligand binding free energy into individual energy terms. The major difference between them is that force-field-based scoring function employs the theoretical framework, including the energy function and the associated parameters, from well-established models; whereas the empirical scoring function usually adopts a flexible functional form which is obtained from regression fit. Several force-field-based methods have introduced empirical parameters to reconcile the contributions of its energy terms just like an empirical scoring function. For example, AutoDock4 (Huey, 2007) calibrated the energy terms using a large set of diverse protein-ligand complexes (see Table 1).

The main drawback of the empirical scoring functions is their dependence on the experimental data set used to perform regression analyses and fitting. This means that based on the limited training set, it is rather difficult to establish a comprehensive and consistent description of all possible weighting factors in protein-ligand binding within the framework of an empirical scoring function. i.e. there is no guarantee that an empirical scoring functions is able to predict the binding affinity of the ligands structurally very different from those used in the training set. Nowadays, researchers are able to use several thousands of protein-ligand complexes at a better quality to calibrate or test their scoring functions. The doubt is largely removed while the remaining concern is that the public experimental data are usually measured under different conditions.

3. Knowledge-Based Scoring Function

Since the pioneer work by Tanaka and Scheraga (Tanaka, 1976), a large number of knowledge-based scoring functions have been developed and widely applied to protein structure prediction and protein-ligand studies. The knowledge-based scoring function is also known as statistical potentials. It employs energy potentials that are derived from the structural information embedded in experimentally determined atomic structures (Miyazawa, 1985; Sippl, 1990). The principle behind knowledge-based scoring functions is simple. The interactions of protein-ligand complexes are modeled using pairwise potentials. The pairwise potentials are directly obtained from the occurrence frequency of atom pairs in a database using the inverse Boltzmann relation. The potentials are computed by

$$S = \sum w(r) \quad (3)$$

$$w(r) = -k_B T \ln[g(r)] = -k_B T \ln \left[\frac{f(r)}{f^*(r)} \right] \quad (4)$$

where $w(r)$ is the pairwise potential between protein and ligand, k_B is the Boltzmann constant, T is the absolute temperature of the system, $f(r)$ is the occurrence frequency of the atom pair at distance r and $f^*(r)$ is the averaged occurrence frequency in the reference state. If an interatomic distance occurs more

often than the average value in the reference state where interatomic interactions are zero, the pairwise potential is assumed as a favorable contact, and vice versa. A challenge in deriving knowledge-based scoring functions is how to achieve the reference state (Thomas, 1996).

The various knowledge-based scoring functions differ between each other in the training sets of protein-ligand complexes used to obtain the potentials, the form of the energy function, the definition of protein and ligand atom types, distance cutoff, and the approximation of the reference state. For example, SMOG uses 13 atom types, two distance intervals and a reference state determined by a self-consistent method (Ishchenko, 2002). SPA uses 22 atom types, 16 distance intervals and a reference state circumvented by Monte Carlo optimization (Yan, 2012). Some methods have combined knowledge-based potentials with solvation and entropy terms (Huang, 2010; Gohlke, 2000; Yan, 2015). In this sense, the boundary between knowledge-based and empirical scoring functions was blurred. The representative knowledge-based scoring functions are SMOG (DeWitte, 1996), PMF (Muegge, 1999), DrugScore (Gohlke, 2000), BLEEP (Mitchell, 1999), ITScore (Huang, 2006), SPA (Yan, 2012).

Unlike force-field-based and empirical scoring functions, knowledge-based potentials attempt to capture all of the energetic factors in protein-ligand interaction implicitly with the pairwise potentials. The major advantage of the knowledge-based scoring functions is their computational simplicity. The deduction of the potentials required for the scoring function only requires the knowledge of a set of protein-ligand complex structures, and such knowledge is relatively rich and still increasing due to the contributions from structural biologists. This overcomes the disadvantage of knowledge-based scoring function that its derivation is based on structural information implicitly encoded in the limited sets of protein-ligand complex structures. In addition, these scoring functions can be easily applied to systems that were not used for the setup of the scoring function, which makes them attractive to apply for very large compound databases.

4. Machine-Learning-Based Scoring Function

Machine-learning-based (ML-based) scoring function is relatively new compared to above three classes of scoring functions since it was first emerged in 2004 (Deng, 2004). These methods employ quantitative structure-activity relationship (QSAR) analysis into protein-ligand interaction evaluation. The machine-learning techniques are applied to derive statistical models that compute protein-ligand binding scores, once the properties of the ligand and the protein as well as their interaction patterns are coded with certain descriptors. The starting point of the method is generally a large pool of descriptors including the descriptors accounting for specific interactions (electrostatic interactions, hydrogen bonds, or aromatic stacking), geometrical descriptors (surface or shape properties), and conventional ligand-based descriptors (molecular weight, number of rotatable bonds, etc.). A variety of machine-learning algorithms, such as neural network, Bayesian classifiers, random forest, and support vector machine, are employed for variable selection. Similar to empirical scoring functions, the method also needs a training set of protein-ligand complexes with known structures and binding data to derive their final models. Typical examples include NNScore (Durrant, 2010), RF-Score (Ballester, 2010), SFCscore^{RF} (Zilian, 2013) and ID-Score (Li, 2013).

ML-based scoring functions have attracted attentions and discussions recently due to their good performance in predicting the binding affinity (Ballester, 2010; Kramer, 2010; Ballester, 2011; Zilian, 2013; Ballester, 2014). ML-based scoring functions achieved Pearson correlation coefficients over 0.700 for the testing benchmark, which is superior to other types of scoring functions (shown in Table 4 in the

next section). However, Gabel et al. argued that the ML-based scoring functions perform very poorly in the docking test (Gabel, 2014). It means ML-based scoring functions are unable to identify the “native” pose out of the docking decoys. The reason could be that the ML-based functions are over-trained on descriptors that do not depict protein-ligand interactions but interaction-independent atom-pair counts which are insensitive to docking poses. Thus, appropriate improvement is needed to address the issue and extensive validations are necessary for the ML-based scoring functions.

5. Consensus Scoring Function.

Consensus scoring function in essence is not a type of scoring function but combines the scores from multiple scoring functions. It is first proposed by Charifson et al. (1999) and become a new trend in this field. The aim of consensus scoring is to balance errors in single scores and improve the probability of identifying “true” modes/binders. Generally, they are employed as a post-processing step after docking runs. The crucial implementation in consensus scoring is the design of an appropriate consensus scoring strategy of individual scores so that the “true” modes/binders can be discriminated from others (Wang, 2001; Clark, 2002). Commonly used consensus scoring strategies include vote-by-number, number-by-number, rank-by-number, average rank, linear combination, etc (Oda, 2006). Representative examples of consensus scoring are MultiScore (Terp, 2001), X-Cscore (Wang, 2002), GFScore (Betzi, 2006), SCS (Teramoto, 2007a), SeleX-CS (Bar-Haim, 2009).

Many studies have showed that employing consensus scoring approaches can improve the performance by compensating for the deficiencies of multiple individual scoring functions and tend to reduce the number of false positives that are identified by individual scoring functions (Bissantz, 2000; Feher, 2006). It is worth noting that the potential advantage of consensus scoring is limited when the predictions from different scoring functions are highly correlated.

Evaluation of Scoring Functions

1. Evaluation Methods

As introduced, there are three applications for the scoring functions, identifying the “native” pose out of the conformational decoys, predicting the binding affinity of the protein-ligand interactions and virtual database screening. In line with these three applications, the performance of a scoring function can be evaluated with three related but independent methods. Researchers have named them as docking power, scoring power and screening power (Li, 2014).

The docking power requires the structural comparisons of the docking decoys with the “native” conformation of the ligand. It determines the performance of a scoring function in identifying the best-scored decoy as the “native” binding mode. Normally, if RMSD value of the best-scored binding pose to the “native” pose is less than a predefined cutoff, it is considered as a successful identification of the “native” pose by the scoring function. However, this criterion could be problematic in some cases due to its simplicity. For example, small or nearly symmetrical ligands are likely to obtain relative small RMSD values even when they are randomly placed in a small active site. On the contrary, large flexible ligands are likely to obtain the large RMSD values. To overcome these limitations, several alternative methods have been proposed for pose evaluations, such as relative displacement error (RDE) (Abagyan, 1997), interaction-based accuracy classification (IBAC) (Kroemer, 2004), real space R-factor (RSR) (Yusuf, 2008), and generally applicable replacement for RMSD (GARD) (Baber, 2009).

The scoring power refers to the prediction accuracy of a scoring function on how closely it can estimate the binding affinities in comparison with the experimental measurements. The predicted scores with scaling by scoring functions usually reproduce the relative values of experimental binding affinities. Therefore, the correlations between the predicted binding scores and experimentally measured binding affinities were generally used to evaluate the accuracy of binding affinity prediction. The higher the correlation coefficient is, the better the scoring power is. The commonly-used correlation is the Pearson correlation (CP) which is computed as

$$C_p = \frac{\sum (S_i - \langle S \rangle)(G_i - \langle G \rangle)}{\sqrt{\sum (S_i - \langle S \rangle)^2} \sqrt{\sum (G_i - \langle G \rangle)^2}} \quad (5)$$

where S_i is the score of a protein-ligand complex computed by a certain scoring function, G_i is the experimental binding affinity. According to the comparative assessment of scoring functions (Li, 2014), the performance of scoring functions in binding affinity is limited with the correlations below 0.70. This could be resulted from two aspects. Firstly, current scoring functions are still in need of improvement as stated in this chapter. Second, the correlation coefficient is dependent on the accuracy and reliability of experimental affinities. Experimental affinities are collected from diverse sources where the binding constants are likely measured under different conditions; these differences can lead to inherent errors among the experimental affinities although carefully cultivated by the researchers (Wang, 2004; Block, 2006; Dunbar Jr, 2011). The inherent errors of the collected experimental binding constant limit the correlation, and make the improvement much harder when correlation coefficient is relatively high.

The screening power determines the ability of a scoring function in discriminating the true binders to a given protein against a large database of compounds. Virtual database screening tests whether or not a scoring function is able to rank the known binders/inhibitors above the inactive compounds in a database. The enrichment test is often taken to quantify the performance of a scoring function in virtual database screening. The enrichment is defined as the accumulated rate of active inhibitors/binders found above a certain percentile of the ranked database that includes the active binders and inactive ligands. A higher enrichment at a fixed percentage of the ranked database can be taken to indicate a better scoring function. Other measurements for virtual database screening are AUC (Area Under the Curve, such as AOC (receiver operating characteristic) curve) and KS (Kolmogorov-Smirno) test, which are more appropriate when the number of inactive ligands is comparable to the number of active binders.

Ideally, an accurate scoring function should be able to perform well above three evaluations. However, most of the existing scoring functions usually perform well on only one or two of the evaluations and fail on others (Li, 2014). In addition, the performances of scoring functions are sometimes sensitive to the testing set. Therefore, to fully evaluate the performance of a scoring function, the above three evaluations are suggested to be carried out with more than one test sets.

2. Evaluation Exercise and Benchmark

An objective and publicly available exercise or benchmark for assessing the performance of the scoring functions can provide practical guidance for the users to make appropriate choices. It can also elucidate the common weakness in the scoring functions for future improvements. The commonly-used exercises

and benchmarks in the research community are CSAR (Community Structure-Activity Resource, <http://www.csardock.org/>) (Smith, 2011) and SAMPL (Statistical Assessment of the Modeling of Proteins and Ligands, <http://www.eyesopen.com/SAMPL>) (Nicholls, 2008), CASF (Comparative Assessment of Scoring Functions, <http://www.pdbbind-cn.org/>) (Cheng, 2009) and DUD (Directory of Useful Decoys, <http://dud.docking.org/>) (Huang, 2006; Irwin, 2008). CSAR and SAMPL exercise results are blind to participants before the exercises were finished. The exercises are normally carried out annually and the benchmarks are often updated and improved with new and higher-quality data. It is worth mentioning that scoring functions are often applied in combination with molecular docking methods to predict ligand binding poses and ligand binding affinities or to identify active compounds through virtual screening. Thus, the evaluation exercises or benchmarks are designed to assess not only the scoring functions but also the docking programs.

3. Comparative Evaluation

By applying the evaluation methods and using the benchmark, researchers have recently compared the performance of different scoring functions testing on the benchmark CASF-2013 (Li, 2014; Yan, 2015). CASF-2013 is composed of 195 protein-ligand complexes with known structures and experimental affinities. To evaluate the scoring functions, a set of binding poses were generated for each protein-ligand complex with RMSDs ranging from 0 to 10 angstroms. The binding poses were scored by the scoring functions including Force-field-based, empirical and knowledge-based scoring functions. The testing results of docking power and scoring power were listed in the Table 3 and Table 4.

It can be seen that all the scoring functions predict the affinities with correlation coefficients lower than 0.7. This suggests that all the testing scoring functions are still not accurate in predicting binding affinities. Compared to the scoring power, more than half scoring functions achieves success rate over 80% in the docking power when top three poses are considered, suggesting that the scoring functions have higher capability in identifying the “native” conformation. Based on the comparative evaluations of different scoring functions, there is no clear advantage for a particular class of scoring function since the performance of the scoring functions in the same class is diversely ranked in the tables. It is worth noting that SPA-SE ranks best for both docking power and scoring power, as stated in the paper (Yan, 2015), this is because SPA-SE incorporates solvation effect in the scoring and binding specificity in the optimization, which are discussed next.

CHALLENGES AND SOLUTIONS

The molecular docking involves concerted performance of sampling and scoring the binding poses. To this end, there are some major challenges, such as protein flexibility in the sampling, implicit or explicit solvation effect of the binding site, entropy contribution to the binding free energy, covalent bonding between the protein and the ligand, metal ion coordination during binding, binding specificity of molecular recognition, structural water molecules in the binding site and so on. Most of the challenges are related to both sampling and scoring methods and a few are beyond them. Here, the authors introduced the major challenges on the side of the scoring function and recent solutions, such as incorporating

Table 3. Comparative evaluations of scoring functions in the docking power

Scoring Function	Classification	Success Rate (%)		
		Top One	Top two	Top Three
SPA-SE	Knowledge-based	83.1	89.2	92.8
SPA	Knowledge-based	82.1	88.2	90.8
ChemPLP@GOLD	Empirical	81.0	86.7	89.7
ChemScore@GOLD	Empirical	77.9	83.1	88.2
GlideScore-SP	Empirical	78.5	85.6	87.7
LigScore2@DS	Empirical	76.9	84.1	86.7
PLP1@DS	Empirical	76.9	84.1	86.2
Alpha-HB@MOE	Empirical	75.4	82.6	86.2
GlideScore-XP	Empirical	74.4	82.6	85.6
ASP@GOLD	Knowledge-based	71.8	81.5	87.2
GoldScore@GOLD	Force-field-based	71.3	81.0	85.6
LUDI1@DS	Empirical	59.0	75.4	83.1
Affinity-dG@MOE	Empirical	63.1	74.9	81.0
ChemScore@SYBYL	Empirical	59.5	69.2	75.4
London-dG@MOE	Empirical	59.5	73.8	78.5
X-Score ^{HM}	Empirical	61.0	73.3	77.9
G-Score@SYBYL	Force-field-based	45.1	61.5	72.3
Jain@DS	Empirical	48.2	62.1	70.8
PMF@SYBYL	Knowledge-based	51.8	60.0	66.7
PMF04@DS	Knowledge-based	51.8	62.6	66.2
ASE@MOE	Empirical	51.3	60.0	63.6
D-Score@SYBYL	Force-field-based	18.5	29.7	42.6

The results of scoring function SPA-SE and SPA are obtained from the paper (Yan, 2015) and others are from the paper (Li, 2014). For more details, the readers are suggested to look into the citations.

the implicit or explicit solvation effect of the binding site, incorporating the entropy contribution to the binding free energy and taking account of binding specificity in the development strategy. Scoring functions accounting for such contributions are more accurate than those without these considerations.

Solvation Effect

It is believed that water molecules actively participate in and affect the process of protein-ligand binding in aqueous solvent (Dill, 2005; Levy, 2006; Berne, 2009; Yan, 2015). The hydration shell that forms around the protein and the ligand is of particular importance to their binding activity. During the binding process, both solvated protein and ligand will be partly dehydrated and the water layers will be gradually excluded into the bulk water, making the ligand diffuse into the active site of the protein in the solvent. The role of water molecules not only assists the formation of the protein-ligand interface but also prob-

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Table 4. Comparative evaluations of scoring functions in the scoring power

Scoring Function	Classification	C_p	D_s	N_c
SPA-SE	Knowledge-based	0.662	1.69	195
SPA	Knowledge-based	0.656	1.70	195
X-Score ^{HM}	Empirical	0.614	1.78	195
ChemScore@SYBYL	Empirical	0.592	1.82	195
ChemPLP@GOLD	Empirical	0.579	1.84	195
PLP1@DS	Empirical	0.568	1.86	195
G-Score@SYBYL	Force-field-based	0.558	1.87	195
ASP@GOLD	Knowledge-based	0.556	1.88	195
ASE@MOE	Empirical	0.544	1.89	195
ChemScore@SYBYL	Empirical	0.536	1.90	189
D-Score@SYBYL	Force-field-based	0.526	1.92	195
Alpha-HB@MOE	Empirical	0.511	1.94	195
LUDI3@DS	Empirical	0.487	1.97	195
GoldScore@GOLD	Force-field-based	0.483	1.97	189
Affinity-dG@MOE	Empirical	0.482	1.98	195
LigScore2@DS	Empirical	0.456	2.02	190
GlideScore-SP	Empirical	0.452	2.03	169
Jain@DS	Empirical	0.408	2.05	191
PMF@DS	Knowledge-based	0.364	2.11	194
GlideScore-XP	Empirical	0.277	2.18	164
London-dG@MOE	Empirical	0.242	2.19	195
PMF@SYBYL	Knowledge-based	0.221	2.20	191

The results of scoring function SPA-SE and SPA are obtained from the paper (Yan, 2015) and others are from the paper (Li, 2014). C_p is the Pearson correlation between experimental and predicted affinities computed by Equation 5; the D_s is the standard deviation in the linear correlation; The N_c is the number of complexes used in the Pearson correlation calculation. For more details, the readers are suggested to look into the citations.

ably makes the active site more adaptable for ligand binding than other possible binding sites, smoothing the binding energy landscape (Papoian, 2003; Levy, 2006). In this sense, the solvation effect is assumed to be a regulator to the energetics as well as the specificity of ligand binding.

To account for the solvation effect, much progress has been made in the development of both explicit and implicit solvent models (Feig, 2004; Purisima, 2013). Explicit model with inclusion of water molecules can provide detailed structural and energetic information; whereas it significantly increases computational cost (Levy, 1998). Especially, it is unaffordable to screen a large library of small molecules in the docking-based drug discovery. The charge-dependent implicit solvent models, such as PB/GBSA (Poisson-Boltzmann/Generalized Born Surface Area) models, are widely taken as post-processing methods to compute free energies based on molecular dynamics simulations or energy minimizations (Kollman, 2000; Lindstrom, 2011); whereas, these models are rarely incorporated into the scoring function for the process of protein-ligand docking (Zou, 1999). Alternatively, the charge-independent implicit solvent models have the advantage of less computational cost due to simple additive expression since the models

are based on solvent accessible surface area (SASA) of each atom and the atomic solvation parameter (Eisenberg, 1986). However, the atomic solvation parameters are usually independently optimized (Wang, 2001; Hou, 2002; Guvench, 2004; Hou, 2005), which are not transferable to the scoring functions of protein-ligand interactions which are described with different classifications of atomic types compared to those used in the solvent models.

Given above difficulties, two knowledge-based scoring functions SPA-SE (Yan, 2015) and ITScore (Huang, 2010) have effectively combined statistical atomic-pair potentials and the atomic solvation energy. The atomic solvation energy is represented by the charge-independent implicit solvent model which accounts for the contribution of solvation energy to the protein-ligand binding based on the change of SASA (Solvent Accessible Surface Area) upon binding. SPA-SE optimized the energetic parameters by maximizing the affinity and specificity predictions of ligand binding, while ITScore employed an iterative method to derive the parameters until the predicted energy converges to the experimentally observed one. Both methods have showed improvements in predicting the binding affinity and identifying the “native” pose.

Entropy Contribution

Conformational entropy is an important component to the free energy change upon binding of a ligand to the protein (Finkelstein, 1989; Murray, 2002; Salaniwal, 2007). The entropy contribution to the binding free energy essentially is arisen from the reduction of the translational and rotational degrees of freedom in the ligand and from changes in the flexibility of the protein during binding, as well as the water molecules involved (Hermans, 1997; Amzel, 1997). The computation of the entropic contribution is very time-consuming (Baron, 2009). Therefore, incorporating such effects in an efficient scoring function is still an important challenge in the field of protein-ligand docking. Some attempts to incorporate entropy in scoring functions have been reported in the literatures which implemented them in the re-scoring schemes (Ruvinsky, 2005; Ruvinsky, 2007; Lee, 2008).

Ruvinsky and Kozintsev (2005) proposed a way to estimate contributions of the translational, rotational and torsional entropy to protein-ligand affinity. In their approach, A number of docking runs are carried out and the generated conformations are clustered by structural similarity. The size of each cluster is then measured to estimate the entropic contribution based on the assumption that large clusters of conformations are indicative of favorable entropic contributions of the local energy landscapes. With this approach, testing scoring functions have shown some improvements in identifying the “native” binding pose (Ruvinsky, 2007). Lee and Seok (2008) introduced a probability function to analyze the populations of diverse binding poses in the context of statistical mechanics. The approach estimates the contribution of the state represented by a sampled conformation of the configurational integral (Xiang, 2002). The approach in combination with 11 scoring functions has resulted in improved accuracy in binding mode prediction. Both approaches are relatively low computational cost and need the combination with other pre-existing scoring functions. However, only the entropy contribution resulted from the ligand were accounted for in these studies. Some other MD-based methods were proposed to estimate the conformational entropy of the protein (Andricioae, 2001; Baron, 2006; Baron, 2008), but they are challenging to be widely employed in the scoring functions due to highly computational demanding.

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In addition to the entropic contributions of the ligand and protein, other key components still require our attention. For example, the entropic contribution of the solvent molecules, particularly those at the binding pocket is normally forgotten (Chiba, 2011; Breiten, 2013). The very limited movement of buried waters in the protein-ligand complex and the huge variation arising from desolvation make this an important issue.

Binding Specificity

Biomolecular recognition requires both stability and specificity to realize its function (Janin, 1995; Havranek, 2003; Wang, 2003; Wang, 2007; Kortemme, 2004; Bolon, 2005). The stability of the complex is determined by the binding affinity while the specificity is measured by the discrimination against competitive binding complexes. Most scoring functions of protein-ligand binding, whether force-field based, empirical, knowledge-based scoring functions or other quantitative description of protein-ligand interactions, mainly aim to predict the binding affinities. The strategy of developing these methods seeks to optimize the stability based on the combination of energetics and shape complementarity but without explicit consideration of binding specificity.

Practically, the quantification of binding specificity is a challenge since it requires comparison of all the possible binding complexes. To circumvent the challenge, the concept of intrinsic specificity was proposed to describe the binding specificity (Wang, 2003; Wang, 2007; Yan, 2012). According to the funnel-like energy landscape of protein-ligand binding (Wang, 2003; Wang, 2007; Bryngelson, 1995; Janin, 1996; Rejto, 1996; Miller, 2008; Tsai, 1999; Dominy, 2004; Z. Liu, 2004; Levy, 2004), the intrinsic specificity ratio (ISR) was quantified to measure the binding specificity. With the quantification of the binding specificity, Yan and Wang (Yan, 2012) developed a novel scoring function SPA by optimizing both specificity and affinity. The development strategy for SPA is to simultaneously optimize binding specificity and affinity of ligand binding. SPA was validated by testing the benchmark set and the performance was compared with 16 popular scoring functions. The test results showed that SPA outperformed the compared scoring functions. The work built up a new frame to develop and improve the scoring function of protein-ligand interactions by incorporating the binding specificity.

FUTURE RESEARCH DIRECTIONS

With the efforts in the past three decades, scoring functions of protein-ligand interactions have made much progress and have been used for many successful applications. Currently, there are large data sets of protein-ligand complexes publicly available with known structures and binding affinities to improve the quality. Also, quite a few benchmarks have been created so that different scoring functions can be compared in a more objective manner. This will be undoubtedly further enhancing the improvement of current scoring functions.

Despite the numerous successful applications of molecular docking-based drug discovery, a widely accepted fact is that the major weakness of current computational protein-ligand docking lies not in the sampling methods but in the scoring functions (Kitchen, 2004; Danishuddin, 2015). The benchmark testing suggests that current scoring functions can provide promising results in predicting the correct

ligand binding pose, but their performance is relatively weak in predicting protein-ligand binding affinities. In virtual screening, it is still challenging for scoring functions to differentiate low- to medium-affinity binders from non-binders. Thus, considerable efforts are still needed to the development and improvement of the scoring function aiming to rapidly and accurately quantify the interaction between the protein and the ligand (Huang, 2010; Wang, 2013; Liu, 2015). As discussed, a current trend in the field is to focus on the inclusion of solvation effect, entropy contributions as well as binding specificity. Scoring functions accounting for those are more accurate but are also computationally more expensive. The fast and accurate prediction of a ligand tightly and specifically binding to a target protein will be the final aim of the scoring functions in the docking-based drug discovery and design.

So far, there are a large number of alternatives of the scoring functions, most of them are generic models derived from the large-scale experimental data of protein-ligand complexes and are presumably applicable to all kinds of target classes. However, it is still difficult to develop a scoring function which can perform equally well across many different protein families (Seifert, 2009; Scharfe, 2012; Cheng, 2012). Binding sites and recognition processes have unique features that ultimately render protein-ligand interactions specific, which in turn might require re-calibrating of scoring schemes on a case-by-case basis. Thus, scoring functions can be improved by tailoring them to a specific target site or to multiple related receptors. For example, the kinase family-specific potential of mean force (kinase-PMF) (Xue, 2010) is kinase-specific scoring function adjusted from the original PMF04 (Muegge, 2006). Target-specific scoring functions have shown improved performance and enhanced screening enrichment in the tests (Teramoto, 2007b; Pham, 2008; Seifert, 2008). It is a promising trend to develop target-specific scoring functions in the future since the improvement of generic scoring functions is often limited with the compromise between the accuracy and computational cost.

CONCLUSION

The discovery and development of novel drugs are highly complex, time-consuming and expensive. Docking-based drug discovery and design is gaining prominence as a method to filter the very large chemical space. Molecular docking-based virtual screening becomes routine in both pharmaceutical companies and academic groups for early-stage drug discovery. Scoring function is at the heart of molecular docking by assisting a docking program to efficiently explore the binding space of a ligand. It is also responsible for evaluating the binding affinity once the binding poses are generated. Therefore, the performance of scoring functions has a significant impact on the docking-based drug discovery. A multitude of scoring functions has been reported in the past three decades and new ones are still emerging. The current trend to improve the performance of scoring functions focusing on the inclusion of additional factors to account for solvation and entropic effects, as well as the strategy to quantify the binding specificity and to develop the target-specific scoring functions. As large data sets of protein-ligand complexes with high quality are available to the public, as well as the computation power increasingly improves, researchers will reach better tradeoff between the accuracy and computational cost for the scoring functions in the coming years. It will undoubtedly produce more intriguing and important applications through molecular docking-based drug design and discovery.

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KEY TERMS AND DEFINITIONS

Benchmark: A set of diverse protein-ligand complexes used for testing the scoring functions or docking methods.

Conformational Space: The degrees of freedom involved in the process of placing a ligand onto the protein.

Docking: The process of searching the possible locations for a ligand onto the binding site of a protein.

Sampling: The exploration of the conformational space given by the movements of the docking partners.

Scoring: The measurement of the fitness for the interactions of a ligand with a protein.

Solvation: The process that the water molecules surround the protein and the ligand.

Specificity: The feature of a ligand to recognize its own binding partner and to discriminate against others.

Virtual Screening: Computationally search the library of small molecules to identify the lead compounds which most favorably bind with a target protein.